



NAME:
HABIB ID:
SECTION:
INSTRUCTOR:

LINEAR ALGEBRA

SPRING 2026

QUIZ 14 (29th April 2026)

SECTIONS L1, L2

Max Marks: 10

Time: 12 minutes

Q. 1 Show that if two vectors $A, B \in M_{nn}$ share an eigenvector, then the sum of the eigenvalues is an eigenvalue of the matrix sum.

Q. 2 If $\underline{x}$ is an eigenvector of A corresponding to λ , then prove that λ^3 is the eigenvalue of A^3 corresponding to eigenvector $\underline{x}$.



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SECTIONS L1, L2

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Time: 12 minutes

Q. 1 Show that if two vectors $C, D \in M_{nn}$ share an eigenvector, then the sum of the eigenvalues is an eigenvalue of the matrix sum.

Q. 2 If $\underline{x}$ is an eigenvector of matrix B corresponding to λ , then prove that λ^3 is the eigenvalue of B^3 corresponding to eigenvector $\underline{x}$.



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SOLUTION QUIZ 14

L1 L2

Problem ①: $C, D \in M_{nn}$. Notate the shared eigenvector as $\underline{v}$.

$$\text{Then: } C\underline{v} = \lambda_C \underline{v} \text{ \& } D\underline{v} = \lambda_D \underline{v} \text{ --- ①}$$

$$\begin{aligned} \text{And clearly: } (C+D)\underline{v} &= C\underline{v} + D\underline{v} \text{ (by associative property)} \\ &= \lambda_C \underline{v} + \lambda_D \underline{v} \text{ (by ①)} \\ &= (\lambda_C + \lambda_D)\underline{v} \text{ (by associative property)} \end{aligned}$$

$$\therefore (C+D)\underline{v} = (\lambda_C + \lambda_D)\underline{v}$$

ie $\lambda_C + \lambda_D$ is eigenvalue of $C+D$. (Shown)

Solution: $Ax = \lambda x \rightarrow A^2x = A(\lambda x) = \lambda(Ax) = \lambda(\lambda x) = \lambda^2x$, similarly for $A^3x = \lambda^3x$.



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QUIZ 14 (24th April 2026)

SECTIONS L3, L4, L5

Max Marks: 10

Time: 10 minutes

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then show that the solutions of the characteristic equation of A are represented by

$\lambda = \frac{1}{2}[(a + d) \pm \sqrt{(a - d)^2 + 4bc}]$. Hence or otherwise, find the eigenvalues of $A = \begin{bmatrix} 1 & 0 \\ 1 & -3 \end{bmatrix}$.



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QUIZ 14 (24th April 2026)

SECTIONS L3, L4, L5

Max Marks: 10

Time: 10 minutes

If $A = \begin{bmatrix} w & x \\ y & z \end{bmatrix}$, then show that the solutions of the characteristic equation of A are represented by

$\lambda = \frac{1}{2}[(w + z) \pm \sqrt{(w - z)^2 + 4xy}]$. Hence or otherwise, find the eigenvalues of $A = \begin{bmatrix} 1 & 1 \\ 0 & -2 \end{bmatrix}$.



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SOLUTION QUIZ 14
L3 L4 L5

$$|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} a-\lambda & b \\ c & d-\lambda \end{vmatrix} = 0$$

$$(a-\lambda)(d-\lambda) - bc = 0$$

$$\cancel{a}^2 - \lambda(a+d) + (ad-bc) = 0$$

Solving this using the quadratic formula:

$$\lambda = \frac{(a+d) \pm \sqrt{(a+d)^2 - 4(ad-bc)}}{2}$$

$$\lambda = \frac{1}{2} [(a+d) \pm \sqrt{a^2 + 2ad + d^2 - 4ad + 4bc}]$$

$$\lambda = \frac{1}{2} [(a+d) \pm \sqrt{(a^2 - 2ad + d^2) + 4bc}]$$

$$(*) \quad \lambda = \frac{1}{2} [(a+d) \pm \sqrt{(a-d)^2 + 4bc}] \quad (\text{Proved})$$

Substituting a, b, c, d in equation above, the solutions are:

Eigenvalues for VERSION A = 1, -3

Eigenvalues for VERSION B = 1, -2